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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Sub'ect	DSA-2
Academic Year	1-III Semester 2025-26
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Roll Number	2520090149
Branch	CSIT
Course Outcome	CO-4

Case Study Topic: Content Recommendation Management using Graph Algorithms

Work Done Summary:

In this case study, we implemented Graph Algorithms for managing content recommendations in the Media Vault system. Each movie, web series, or video is represented as a node, and relationships between similar content are represented as edges. Graph traversal techniques were used to recommend relevant content to users based on their viewing history.

The system analyzed user preferences and content similarities to generate personalized recommendations. Breadth First Search (BFS) and Depth First Search (DFS) algorithms were used to explore connected content efficiently and improve user engagement.

Implementation Details:

1) Created a graph structure to represent digital content relationships.
2) Added nodes for movies, series, and videos.
3) Connected similar content using graph edges.
4) Implemented BFS for recommendation generation.
5) Implemented DFS for content exploration.
6) Analyzed recommendation paths and user preferences.
7) Verified efficient traversal of content networks.

Result: Screenshots / Output

```java
import java.util.*;
public class MediaVaultGraph {
    private int vertices;
    private LinkedList<Integer>[] adjList
    MediaVaultGraph(int v) {
        vertices = v;
        adjList = new LinkedList[v];
        for (int i = 0; i < v; i++) {
            adjList[i] = new LinkedList<>();
        }
    }
    void addEdge(int source, int destination) {
        adjList[source].add(destination);
        adjList[destination].add(source);
    }
    void BFS(int start) {
        boolean[] visited = new boolean[vertices];
        Queue<Integer> queue = new LinkedList<>();
        visited[start] = true;
        queue.add(start);
        System.out.println("Recommended Content Order:");
        while (!queue.isEmpty()) {
            int node = queue.poll();
            System.out.print(node + " ");
            for (int neighbor : adjList[node]) {
                if (!visited[neighbor]) {
                    visited[neighbor] = true;
                    queue.add(neighbor);
```

```
            }
        }
        public static void main(String[] args) {
            MediaVaultGraph graph = new MediaVaultGraph(6);
            graph.addEdge(0, 1);
            graph.addEdge(0, 2);
            graph.addEdge(1, 3);
            graph.addEdge(2, 4);
            graph.addEdge(4, 5);
            System.out.println("Media Vault Content Recommen
            graph.BFS(0);
        }
    }
```

Output:-

```
Media Vault Content Recommendation System
Recommended Content Order:
0 1 2 3 4 5
```

.
The Graph-based recommendation system successfully suggested related content to users. BFS and DFS enabled efficient navigation through content relationships, improving recommendation accuracy and user experience.

GitHub Link: https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system/blob/main/media%20vauly%20graph%20c04

Student Signature   Faculty Signature